QUERIES:

1. CREATE DATABASES:
   mysql> CREATE DATABASE SecurityDB1;
   Query OK, 1 row affected (0.02 sec)

   mysql> CREATE DATABASE SecurityDB2;
   Query OK, 1 row affected (0.00 sec)

2. TABLE 1 (IN SecurityDB1) → USER CREDENTIALS WITH AES ENCRYPTION:
   mysql> USE SecurityDB1;
   Database changed mysql>
   mysql> CREATE TABLE Users (
   -> user_id INT AUTO_INCREMENT PRIMARY KEY,
   -> username VARCHAR(100),
   -> password_encrypted VARBINARY(255),
   -> secret_key VARCHAR(50)
   ->);
   Query OK, 0 rows affected (0.03 sec)

3. TABLE 2 (in SecurityDB2) → MESSAGES WITH AES ENCRYPTION + HASHING:
   mysql> USE SecurityDB2;
   Database changed

```
mysql> mysql> CREATE TABLE
Messages (
   ->    msg_id INT AUTO_INCREMENT PRIMARY KEY,
   ->    original_message VARCHAR(255),
   ->    encrypted_message VARBINARY(255),
   ->    decrypted_message VARCHAR(255),
   ->    hash_value VARCHAR(255),
   ->    base64_encoded TEXT
   -> );
Query OK, 0 rows affected (0.02 sec)
```

4. INSERT ENCRYPTED DATA INTO USERS TABLE:

```
mysql> USE SecurityDB1; Database changed mysql> mysql> -- Insert
user with encrypted password mysql> INSERT INTO Users (username,
password_encrypted, secret_key)
   -> VALUES (
   ->    'Allen',
   ->    AES_ENCRYPT('mySecretPass123', 'encryption_key_123'),
   ->    'encryption_key_123'
   -> );
Query OK, 1 row affected, 1 warning (0.01 sec)

mysql> mysql> -- Check stored
encrypted data
mysql> SELECT user_id, username, HEX(password_encrypted) AS
Encrypted_Password
   -> FROM Users;
+---------+----------+----------------------------------+
| user_id | username | Encrypted_Password               |
+---------+----------+----------------------------------+
|       1 | Allen    | 18239F770FC516B8072F489A2AC391D4 |
+---------+----------+----------------------------------+
1 row in set (0.00 sec)
```

5. RETRIEVE AND DECRYPT PASSWORD:

```
mysql> --
Decrypt the password mysql> SELECT username,
   ->       AES_DECRYPT(password_encrypted, secret_key) AS
Decrypted_Password
   -> FROM Users;
+----------+----------------------------------------+
| username | Decrypted_Password                     |
+----------+----------------------------------------+
| Allen    | 0x6D795365637265745061737331323 3      | +----------+-----------------
---------------------+
1 row in set (0.00 sec)
```

6. INSERT MESSAGES WITH ENCRYPTION, DECRYPTION, HASHING, AND BASE64:

```
mysql> USE SecurityDB2; Database changed mysql> mysql> -- Insert message with AES encryption, hash, and Base64 encoding mysql> INSERT INTO Messages (original_message, encrypted_message, decrypted_message, hash_value, base64_encoded)
    -> VALUES (
    ->     'Secret Message',
    ->     AES_ENCRYPT('Secret Message', 'key123'),
    ->     AES_DECRYPT(AES_ENCRYPT('Secret Message', 'key123'), 'key123'),
    ->     SHA2('Secret Message', 256),
    ->     TO_BASE64('Secret Message')
    -> );
Query OK, 1 row affected (0.01 sec)
```

7. RETRIEVE DATA WITH OPERATIONS:

```
mysql> -- Retrieve all details mysql>
SELECT msg_id,
    ->        original_message,
    ->        HEX(encrypted_message) AS Encrypted_Hex,
    ->        decrypted_message,
    ->        hash_value,
    ->        base64_encoded,
    ->        FROM_BASE64(base64_encoded) AS Base64_Decoded
    -> FROM Messages;
+--------+-----------------+----------------------------------+------------------+-----------------------------------------------------------------+--------------------+--------------------+
| msg_id | original_message | Encrypted_Hex                    | decrypted_message | hash_value                                                      | base64_encoded     | Base64_Decoded     |
+--------+-----------------+----------------------------------+------------------+-----------------------------------------------------------------+--------------------+--------------------+
|      1 | Secret Message   | 9E5050843695055524E47AFFDDF26766 | Secret Message   | 0c48bdfc98b783b5d8e0f550c8296b3dcc685096109f4cbaf42bf9f3d815de21 | U2VjcmV0IE1lc3NhZ2U= | 0x536563726574204D657373616765 |
+--------+-----------------+----------------------------------+------------------+-----------------------------------------------------------------+--------------------+--------------------+
1 row in set (0.00 sec)
```

8. MD5 HASHING (FOR PASSWORDS):

```
mysql> SELECT MD5('Secret Message') AS MD5_Hash;
+----------------------------------+ |
MD5_Hash                          |
+----------------------------------+
| 54b0581b96c2e04197a9f4f50f1e64da |
+----------------------------------+
1 row in set (0.00 sec)
```

9. SHA2 HASHING:

```
mysql> SELECT SHA2('Secret Message', 512) AS SHA512_Hash;
+-------------------------------------------------------------------------------------------------------------------
--------------------------------+
| SHA512_Hash
|
+-------------------------------------------------------------------------------------------------------------------
--------------------------------+
|
0c36e2f06209dd15596988e2269397f4e6272f08b792ed6bf7ae3196fe766d65d047
4264c6da2614f27f3be3671ad32bc5e5b656a73d984316e59c49eed07ba8 |
+-------------------------------------------------------------------------------------------------------------------
--------------------------------+
1 row in set (0.00 sec)
```

10. ENCODING/DECODING:

```
mysql> SELECT TO_BASE64('Confidential Data') AS Encoded,
    ->        FROM_BASE64(TO_BASE64('Confidential Data')) AS Decoded;
+-------------------------+-----------------------------------+ |
Encoded                   | Decoded                           |
+-------------------------+-----------------------------------+
| Q29uZmlkZW50aWFsIERhdGE= |
0x436F6E666964656E7469616C2044617461 |
+-------------------------+-----------------------------------+
1 row in set (0.00 sec)
```